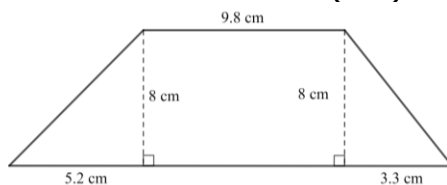


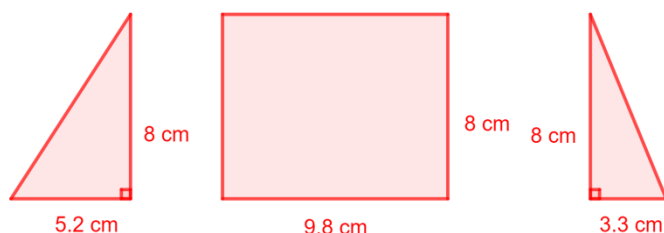
Grade 6
Unit 8
Lesson 3 Practice

Name _____
Date _____
Period _____

A quadrilateral with measurements in centimeters (cm) is shown for questions 1-3.



1. Decompose the quadrilateral into triangles and a rectangle. Sketch the decomposed shapes here.



2. Using the measurements given in the original image, label the base and height of each triangle and the rectangle.

3. Find the area of the original quadrilateral by finding the sum of the areas of the decomposed triangles and rectangle. Include units.

Triangle 1 $\rightarrow 5.2 \cdot 8 \cdot 0.5 = 20.8 \text{ cm}^2$

Total Area $= 20.8 + 78.4 + 13.2$

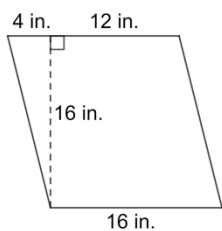
Rectangle $\rightarrow 9.8 \cdot 8 = 78.4 \text{ cm}^2$

$= 112.4 \text{ cm}^2$

Triangle 2 $\rightarrow 3.3 \cdot 8 \cdot 0.5 = 13.2 \text{ cm}^2$

Decompose each quadrilateral to find the area. Lengths are shown in inches (in.).

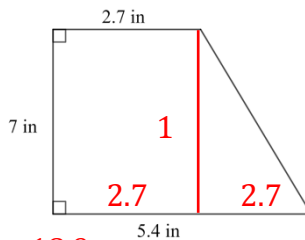
4.



$16 \cdot 16 = 256$

Area = 256 in²

5.



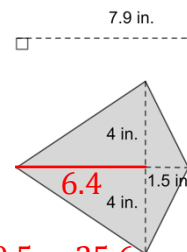
$2.7 \cdot 7 = 18.9$

$2.7 \cdot 7 \cdot 0.5 = 9.45$

$18.9 + 9.45 = 28.35$

Area = 28.35 in²

6.



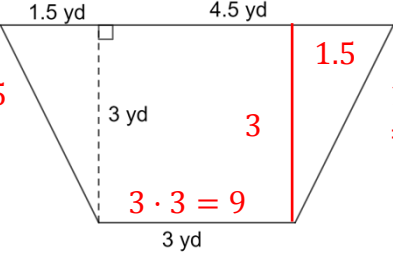
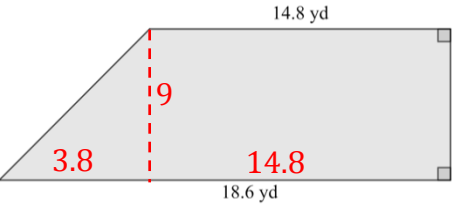
$6.4 \cdot 8 \cdot 0.5 = 25.6$

$1.5 \cdot 8 \cdot 0.5 = 6$

$25.6 + 6 = 31.6$

Area = 31.6 in²

Decompose each quadrilateral to find the area. Lengths are shown in yards (yd).

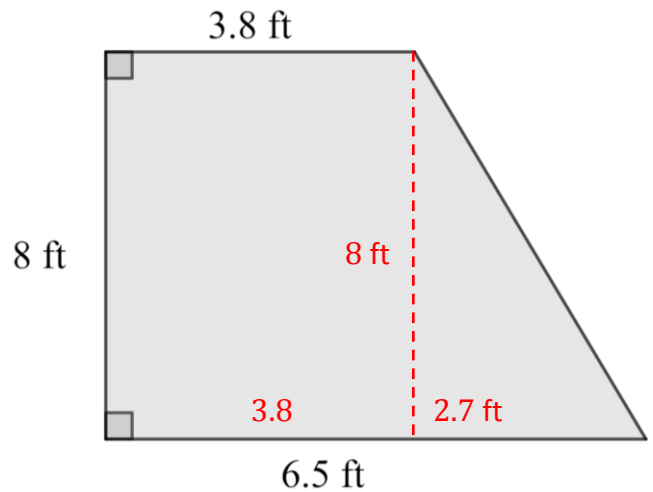
7.  $1.5 \cdot 3 \cdot 0.5 = 2.25$ $3 \cdot 3 = 9$ $1.5 \cdot 3 \cdot 0.5 = 2.25$ $2.25 + 9 + 2.25 = 13.5$	8.  $3.8 \cdot 9 \cdot 0.5 = 17.1$ $14.8 \cdot 9 = 133.2$ $17.1 + 133.2 = 150.3$
Area = <u>13.5 yd²</u>	Area = <u>150.3 yd²</u>

Geri made a sketch of a wall she needs to paint. The sketch of the wall is shown with measurements in feet (ft) for questions 9-10.

9. The home improvement store is offering a sale on their quarts of paint. A quart of paint covers 100 square feet of wall. Will Geri have enough paint if she buys one quart?

$$\begin{aligned}
 8 \cdot 3.8 &= 30.4 \\
 8 \cdot 2.7 \cdot 0.5 &= 10.8 \\
 30.4 + 10.8 &= 41.2 \text{ ft}^2
 \end{aligned}$$

Geri will have enough paint because the area of her wall is 41.2 ft².



10. Determine the amount of paint, in square feet, Geri is short or has left over.

$$\begin{array}{r}
 100.0 \\
 -41.2 \\
 \hline
 58.8
 \end{array}$$

Geri will have 58.8 ft² of paint left over.